PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
IRAMBA DISTRICT COUNCIL
FORM FOUR
BASIC MATHEMATICS

TIME: 3: 00 HOURS

Instructions

1 This paper consists of section A and B with fourteen **(14)** questions.

2 Answer all questions.

3 Each question in section A caries **six (06)** marks while each question in section B carries **ten (10)** marks.

4 All necessary working and answers for each question must be shown clearly.

5 Mathematical tables and non – programmable calculators may be used.

6 All communication devices and any unauthorized materials are **not** allowed in the examination room

7 Write your **Examination number** on every page of your answer booklet(s).

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SECTION A (60 MARKS)

1. (a) Simplify the sum of 85% of 9861 and $\frac{3}{5}$ of 12458. Write your answer correct to two significant figures.

   (b) (i) Mr. Tumai Distributed Tshs 960,000/= awards to students who passed well in their examinations and their respective teachers as follow: 23% to all students who passed Arts Subjects. 15% to students who passed Mathematics and 27% to students who passed science subjects. The remained amount was distributed to teachers. Find the amount that were awarded to teachers.

   (ii) Three bells commence tolling together and toll at intervals of 8, 10 and 12 seconds respectively. How many times do they toll together in 50 minutes?

2. (a) If $a = \sqrt{5},\ b = \sqrt{3}\ , and\ c = \sqrt{2}$. Simplify $\frac{a-b}{b+c}$

   (b) Determine the value of x $\log_5(x+1) - 1 = \log_5(x-3)$

3. In a certain class MSUVA displayed counting numbers less than 30. From the displayed numbers, OKWI mentioned all numbers divisible by 2 and FEITOTO mentioned all numbers that were the multiples of 3.

   (a) (i) Outline all numbers that were mentioned by OKWI and FEITOTO in common.

   (ii) How many numbers were mentioned by either OKWI or FEITOTO?

   (b) If a number was selected at random from the numbers displayed by MSUVA, what is the probability that a selected number was a multiple of 3?

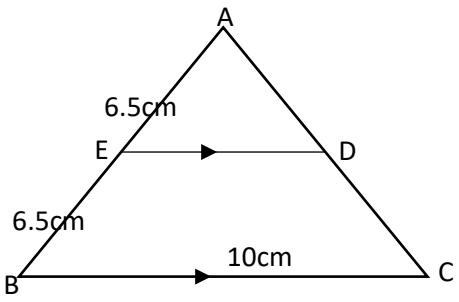
4. (a) Given that $\underline{a}$= 4i + 3j and $\underline{b}$= 6i – 3j, find the value of "h" and "k" if h$\underline{a}$ + k$\underline{b}$ = 10i + j

   (b) A perpendicular line from the point *P(2,-4*) to the line meets the line at point *Z(-1,3).* Find:

   i. Distance $\overline{PZ}$

   ii. If the point *Z(-1,3)* is a mid-point of the line $\overline{PQ}$, find the coordinates of point Q

5. (a) In the following figure; $\overline{ED}//\overline{BC}$ and $\overline{AB} = \overline{AC}$; If $\overline{AB} = 13cm$ and $\overline{BC} = 10cm$, find the Area of BCDE



(b) By considering the Alternating opposite angles theorem, draw the diagram hence identify the corresponding angles.

6. (a) The time "t" taken to buy fuel at XYZ PETROL STATION in Morogoro varies directly as the number of vehicle "V" in the queue and inversely as the number of pumps "P" available in the station. In petrol station with 10 pumps it takes 20 minutes to fuel 40 vehicles.
   i. Write vehicle "V" in terms of pumps "P" and time "t"
   ii. Find the time it will take to fuel 60 vehicles in a station with 4 pumps.

   (b) Mayele bought 3 bottles of juice of capacity 350 ml and Dialo bought 1 bottle of juice of capacity 1 litre.
      i. Who had more juice to drink?
      ii. How much more?
7. (a) A company bought two cars for Tshs 25,000,000/= each. If one car was sold at a profit of 18% and another was sold at a loss of 6%. In the whole transactions there were no loss. What was the profit made by a company?

   (b) What do the following terms mean as used in Accounts?
      (i) Cash book
      (ii) Assets
      (iii) Credit transactions
8. (a) A BODA BODA driver rates after each kilometre. The fare is Tshs. 1000/= for the first kilometre and raise by Tshs. 500/= for each additional kilometre. If Willson want to travel 10 kilometres by BODA BODA. What will he be charged by BODA BODA driver?

   (b) Given that, the last term of Geometric sequence 3, 9, 27 … is 19683. Determine the number of terms
9. (a) From the top of a tower of height 60m the angles of depression of the top and the bottom of a building are observed to be 30^0 and 60^0 respectively. Find the height of the building.

   (b) If the square of the hypotenuse of an isosceles right-angled triangle is 128 cm^2 find the length of each other sides

10. (a) Find $x - y$; Given that $x^2 - y^2 = 24$ and $x + y = 8$

(b) A large rectangular garden in a park is 120m wide and 150m long. A contractor is called in to add a brick walkway to surround this garden by the same width. If the area of the walkway is 2800m^2, how wide is the walkway?

SECTION B (40 MARKS)

11. (a) In certain research the data were summarized as shown in a table below:

Class Mark	10	15	20	25	30	35
Frequency	3	2	10	5	4	1

By using the data above reconstruct a frequency distribution table including class interval and frequency.

(b) Prove that equal chords of a circle subtend equal angles at a centre.

12. (a) A pyramid with vertex V and edges VA, VB, VC, VD each 15cm long has a rectangular base ABCD where AB = CD = 10cm and AD = BC = 8cm.

i. Sketch the pyramid using the above information.

ii. Calculate the height "VO" of the pyramid where "O" is the centre of the rectangle.

iii. Calculate the angle between the base and edge

(b) Two towns P and Q on the Latitude 48^0 are 370 Nautical miles apart. Find the difference in their longitude.

13. (a). (i). Find the possible values of x if the matrix $\begin{pmatrix} x+1 & 9 \\ 3 & x+5 \end{pmatrix}$ has no inverse.

(ii). By using matrix method, solve the equations $\begin{cases} 4x - 9 = -3y \\ 5x - 17 - 2y = 0 \end{cases}$

(b). A transformation is given $x' = 2x + 5y$ and $y' = -x$, find the image of the equation 2x – 3y = 6 under the transformation matrix which performs the above transformations.

14. (a) Given the function $f(x) = x^2 - 2x - 3$; find

i. turning point

ii. x – symmetry

iii. domain and range

iv. maximum or minimum value

(b) A school is preparing a trip for 400 students. The company who is proving the

transportation has 10 buses of 50 seats each and 8 buses of 40 seats, but only 9 drivers available. The rental cost for a large bus is 800,000/= and 600,000/= for the small bus. How many buses of each type should be used for the trip for the least possible cost?